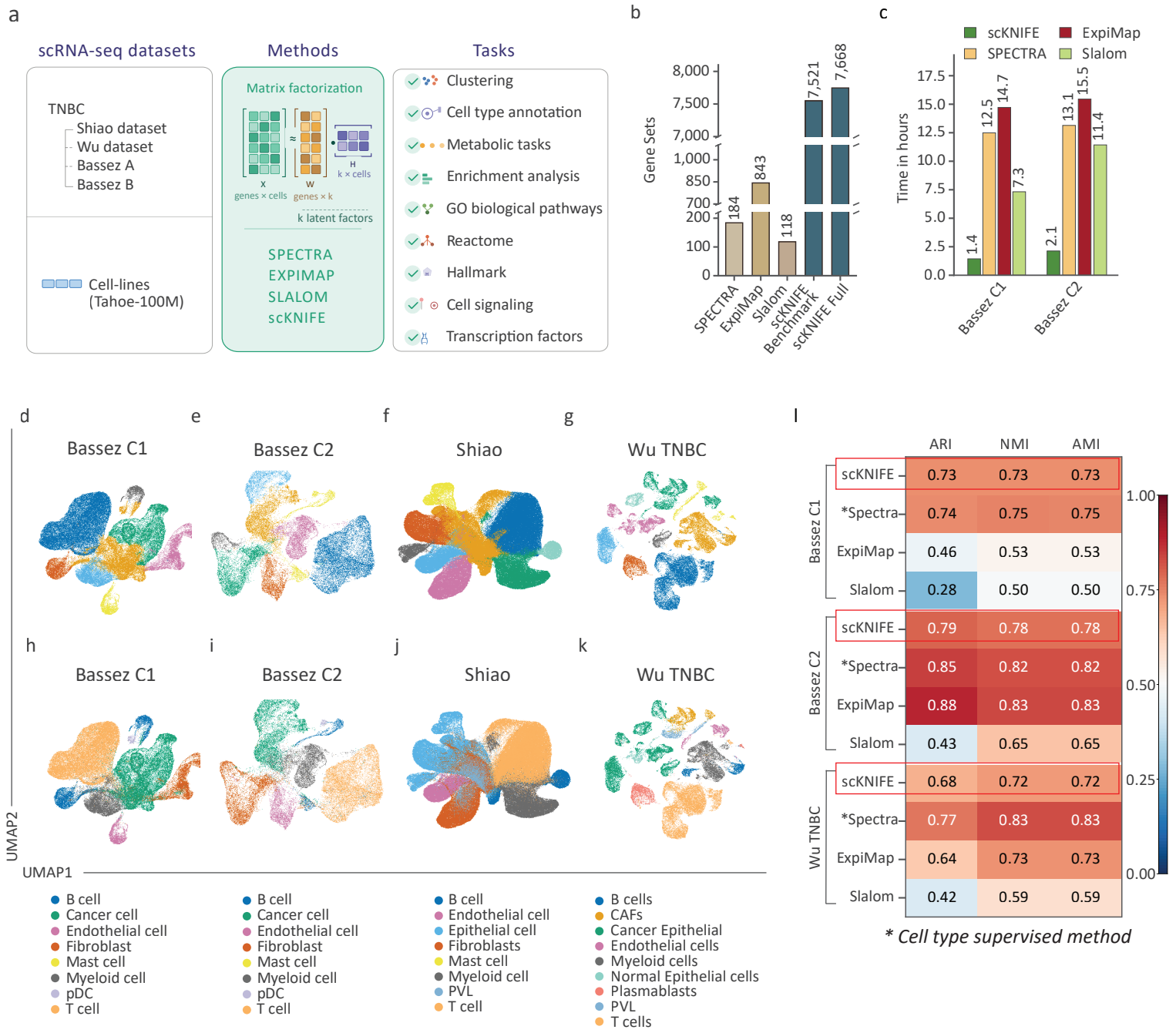

a
scRNA-seq datasets
TNBC
Shiao dataset
Wu dataset
Bassez A
Bassez B
Cell-lines (Tahoe-100M)
Methods
Matrix factorization
x
genes × cells
w
genes × k
H
k × cells
k latent factors
SPECTRA
EXPIMAP
SLALOM
scKNIFE
Tasks
Clustering
Cell type annotation
Metabolic tasks
Enrichment analysis
GO biological pathways
Reactome
Hallmark
Cell signaling
Transcription factors
b
Gene Sets
8,000
7,500
7,000
1,000
850
700
200
100
0
184
843
118
7,521
7,668
SPECTRA
ExpiMap
Slalom
scKNIFE Benchmark
scKNIFE Full
c
scKNIFE
ExpiMap
SPECTRA
Slalom
Time in hours
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
1.4
12.5
14.7
7.3
2.1
13.1
15.5
11.4
Bassez C1
Bassez C2
d
Bassez C1
e
Bassez C2
f
Shiao
g
Wu TNBC
h
Bassez C1
i
Bassez C2
j
Shiao
k
Wu TNBC
UMAP2
UMAP1
B cell
Cancer cell
Endothelial cell
Fibroblast
Mast cell
Myeloid cell
pDC
T cell
B cell
Cancer cell
Endothelial cell
Fibroblast
Mast cell
Myeloid cell
pDC
T cell
B cell
Endothelial cell
Epithelial cell
Fibroblasts
Mast cell
Myeloid cell
PVL
T cell
B cells
CAFs
Cancer Epithelial
Endothelial cells
Myeloid cells
Normal Epithelial cells
Plasmablasts
PVL
T cells
l
ARI
NMI
AMI
Bassez C1
scKNIFE 0.73 0.73 0.73
*Spectra 0.74 0.75 0.75
ExpiMap 0.46 0.53 0.53
Slalom 0.28 0.50 0.50
Bassez C2
scKNIFE 0.79 0.78 0.78
*Spectra 0.85 0.82 0.82
ExpiMap 0.88 0.83 0.83
Slalom 0.43 0.65 0.65
Wu TNBC
scKNIFE 0.68 0.72 0.72
*Spectra 0.77 0.83 0.83
ExpiMap 0.64 0.73 0.73
Slalom 0.42 0.59 0.59
1.00
0.75
0.50
0.25
0.00
* Cell type supervised method
m
Hallmark: PI3K / AKT / mTOR Signaling
scKNIFE Score
r = 0.76
ρ = 0.70
PIK3R1 expression
n
Hallmark: PI3K / AKT / mTOR Signaling
Scanpy Score
r = 0.74
ρ = 0.70
PIK3R1 expression
o
Hallmark: Apoptosis
scKNIFE Score
r = 0.65
ρ = 0.54
TP53 expression
p
Hallmark: Apoptosis
Scanpy Score
r = 0.63
ρ = 0.40
TP53 expression

Figure 2